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Questions

Q1.

Airbags protect occupants by inflating when a car crashes.

Airbags rely on chemical reactions to produce large volumes of gases quickly.
In some airbags, solid sodium azide (NaN_3) decomposes forming nitrogen gas and sodium as the only products.

(a) Write an equation for the decomposition of sodium azide.

State symbols are not required.

(1)

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(b) A passenger airbag requires 120 dm^3 of gas to fill it.

Calculate, using the ideal gas equation, the mass of sodium azide required to fill a passenger airbag in this reaction under standard conditions (101 000 Pa, 25°C).
Give your answer to an appropriate number of significant figures.

$[pV = nRT \qquad R = 8.31\ JK^{-1}\ mol^{-1}]$

(6)

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(c) Two further reactions take place in the airbag.

Reaction **A** $10Na + 2KNO_3 \rightarrow K_2O + 5Na_2O + N_2$

Reaction **B** $K_2O + Na_2O + SiO_2 \rightarrow Na_2K_2SiO_4$

(i) Reaction **A** produces more nitrogen to inflate the airbag.

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Calculate the atom economy, by mass, for the production of nitrogen in reaction **A**
Give your answer to an appropriate number of significant figures.

. **(3)**

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(ii) State the type of reaction taking place in reaction **B**.

(1)

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(Total for question = 11 marks)

Q2.

How many **atoms** are there in 36.0 g of water?

[Avogadro constant = 6.02×10^{23} mol^{-1}]

(1)

- ☒ **A** 3.010×10^{23}
- ☒ **B** 1.204×10^{24}
- ☒ **C** 2.408×10^{24}
- ☒ **D** 3.612×10^{24}

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(Total for question = 1 mark)

Q3.

What is the atom economy, by mass, for the formation of iron in this reaction?

$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$$

[A_r values: C = 12.0 O = 16.0 Fe = 55.8]

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(1)

- ☒ **A** 29.7 %
- ☒ **B** 45.8 %
- ☒ **C** 55.9 %
- ☒ **D** 71.7 %

(Total for question = 1 mark)

Q4.

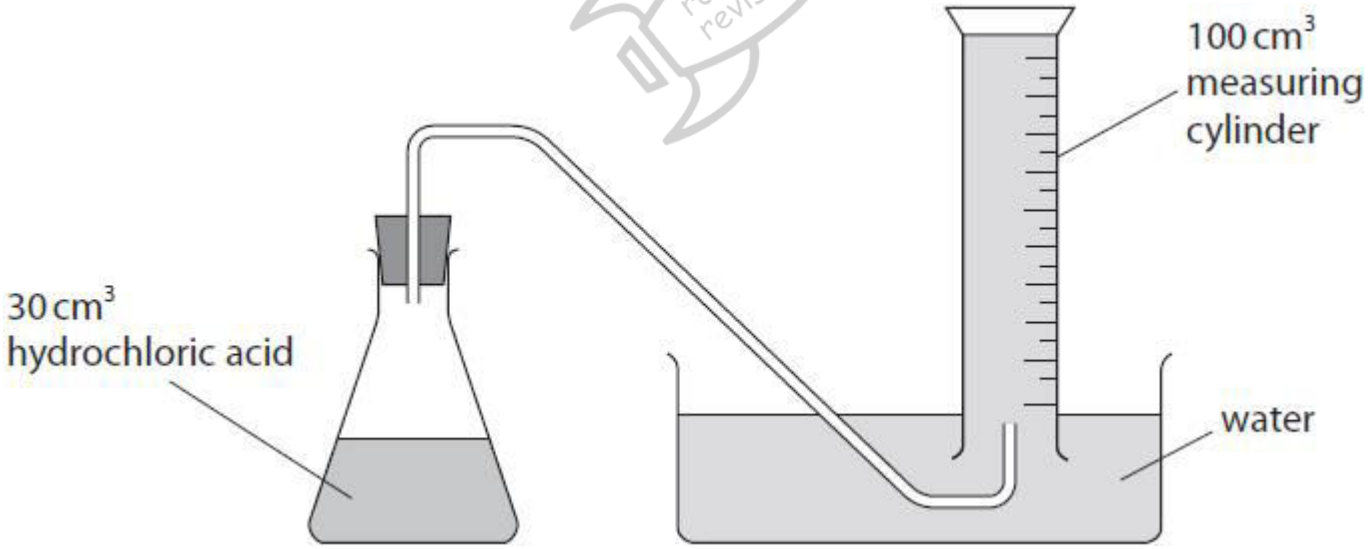
Magnesium carbonate powder reacts with hydrochloric acid.

(a) Complete the equation for this reaction by adding state symbols.

(1)

$$MgCO_3(\quad\quad) + 2HCl(\quad\quad) \rightarrow MgCl_2(\quad\quad) + H_2O(\quad\quad) + CO_2(\quad\quad)$$

(b) A student carried out an investigation to determine the molar volume of carbon dioxide using this apparatus.



The student carried out five experiments, adding a different mass of magnesium carbonate each time. The results are shown in the table.

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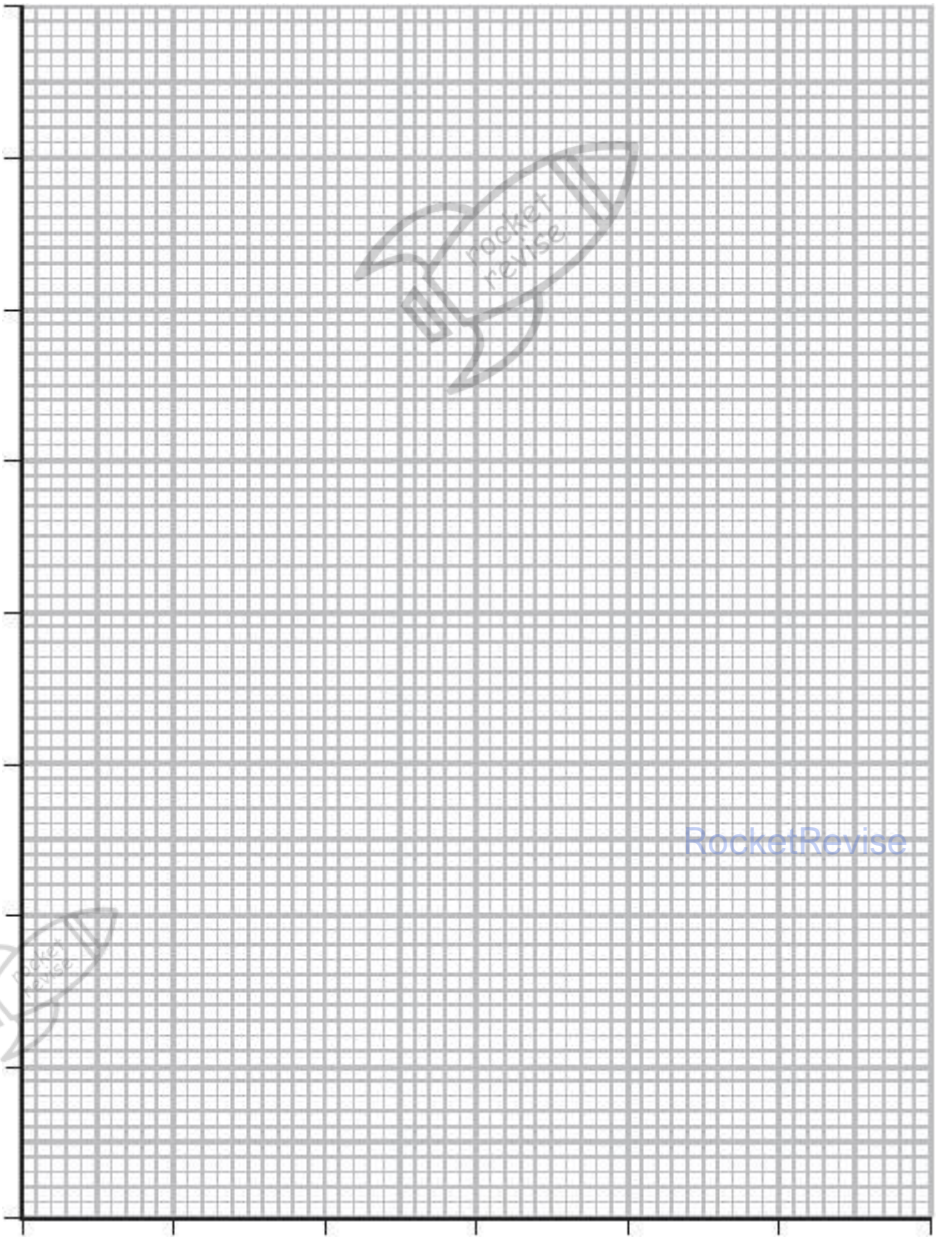
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Mass of magnesium carbonate / g	Volume of gas collected / cm^3
0.05	11
0.10	27
0.15	38
0.20	54
0.25	63

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(i) Plot a graph of these results.

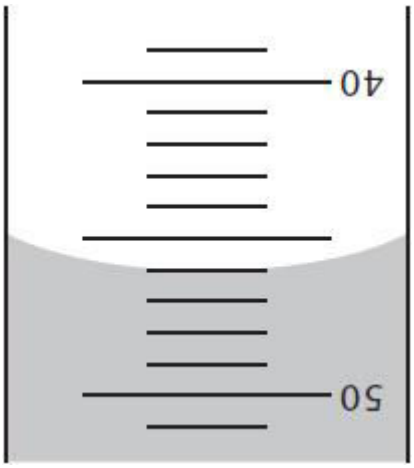
(3)



(ii) A student carried out a further experiment using a different mass of magnesium carbonate.

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Give the volume of gas collected using the **inverted** measuring cylinder.

(1)

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(iii) Determine the mass of magnesium carbonate added in the experiment in (b)(ii), using your graph.

(1)

.....

(iv) Calculate the molar volume of carbon dioxide using your answers to parts (b)(ii) and (b)(iii).
Give your value to an appropriate number of significant figures and include units.
[A_r values: Mg = 24.3 C = 12.0 O = 16.0]

(4)

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(v) The acid must be in excess for each experiment.
Calculate the **minimum** concentration of hydrochloric acid needed for 30 cm^3 of acid to completely react with 0.25 g of magnesium carbonate.

$$MgCO_3 + 2HCl + MgCl_2 + H_2O + CO_2$$

(2)

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(c) The value of molar volume calculated in (b)(iv) was lower than the student expected.

Give **two** reasons for the value being lower than expected.
Assume that the correct amounts of hydrochloric acid and magnesium carbonate were used.

(2)

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(Total for question = 14 marks)

Q5.

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A 2 kg sample of water contains 40 parts per million (ppm) by mass of nitrate ions.

What is the mass, in g, of nitrate ions in this sample?

(1)

- ☒ **A** 8×10^{-2}
- ☒ **B** 5×10^{-5}
- ☒ **C** 8×10^{-5}
- ☒ **D** 5×10^{-8}

(Total for question = 1 mark)

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Q6.

Propene, C_3H_6, is produced in the dehydration of propanol.

$$C_3H_7OH \rightarrow C_3H_6 + H_2O$$

What is the mass, in g, of propene formed from 3.42 g of propanol when the yield is 85.2 % ?

[Relative molecular masses (M_r): $C_3H_7OH = 60$ $C_3H_6 = 42$]

(1)

- ☒ **A** 2.04
- ☒ **B** 2.39
- ☒ **C** 2.91
- ☒ **D** 4.16

(Total for question = 1 mark)

Q7.

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What is the relative formula mass of hydrated ammonium iron(II) sulfate, $(NH_4)_2Fe(SO_4)_2.6H_2O$?

[Relative atomic masses (A_r): H = 1.0 N = 14.0 O = 16.0 S = 32.1 Fe = 55.8]

(1)

- ☒ **A** 284
- ☒ **B** 302
- ☒ **C** 312
- ☒ **D** 392

(Total for question = 1 mark)

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Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In 1.31 g of a chloride of titanium, there is 0.528 g of titanium.

What is the empirical formula of this titanium chloride?

[A_r values: Ti = 47.9 Cl = 35.5]

- ☒ **A** TiCl
- ☒ **B** $TiCl_2$
- ☒ **C** $TiCl_3$
- ☒ **D** Ti_2Cl_4

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(Total for question = 1 mark)

Q9.

A sample of hydrated calcium sulfate, $CaSO_4.\mathbf{x}H_2O$, was heated to constant mass. 3.405 g of anhydrous calcium sulfate and 0.900 g of water were formed.

What is the value of **x**?

[Relative formula mass: $CaSO_4$ = 136.2]

(1)

- ☒ **A** 0.5
- ☒ **B** 2
- ☒ **C** 3
- ☒ **D** 4

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(Total for question = 1 mark)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the percentage atom economy, by mass, for the production of hydrogen in the reaction shown?

$$CH_4 + H_2O \rightarrow CO + 3H_2$$

[A_r values: H = 1.00 C = 12.0 O = 16.0]

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- ☒ **A** 8.8%
- ☒ **B** 17.6%
- ☒ **C** 21.4%
- ☒ **D** 82.4%

(Total for question = 1 mark)

Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A compound has M_r = 84 and its composition by mass is 71.4% carbon, 9.6% hydrogen and 19.0% oxygen.

What is the molecular formula of this compound?

[A_r values: H = 1.0 C = 12.0 O = 16.0]

- ☒ **A** $C_4H_3O_2$
- ☒ **B** $C_4H_4O_2$
- ☒ **C** C_5H_8O
- ☒ **D** $C_6H_{10}O$

(Total for question = 1 mark)

Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

How many molecules are there in 44.0 g of carbon monoxide?

[Avogadro constant $L = 6.02 \times 10^{23}\ mol^{-1}$
A_r values: C = 12.0 O = 16.0]

- ☒ **A** 3.83×10^{23}
- ☒ **B** 6.02×10^{23}
- ☒ **C** 9.46×10^{23}
- ☒ **D** 1.89×10^{24}

(Total for question = 1 mark)

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Q13.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which conversion has the **lowest** percentage atom economy (by mass) for the formation of $CaCl_2$?

☐ **A** $Ca + Cl_2 \rightarrow CaCl_2$

☐ **B** $Ca + 2HCl \rightarrow CaCl_2 + H_2$

☐ **C** $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

☐ **D** $CaCO_3 + 2NaCl \rightarrow CaCl_2 + Na_2CO_3$

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(Total for question = 1 mark)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Propane burns completely in oxygen as shown.

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$$

100 cm^3 of propane was mixed with 600 cm^3 of oxygen and the mixture was ignited.

What is the **total** volume, in cm^3, of the gas mixture at the end of the reaction?
All gas volumes were measured at room temperature and pressure.

☐ **A** 300
☐ **B** 400
☐ **C** 700
☐ **D** 800

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(Total for question = 1 mark)

Q15.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

There are 6.02×10^{23} atoms in 0.25 mol of

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☒ **A** He

☒ **B** H_2O

☒ **C** BH_3

☒ **D** CH_4

(Total for question = 1 mark)

Q16.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the relative formula mass of hydrated sodium carbonate, $Na_2CO_3.\ 10H_2O$?

[A_r values: H = 1.0 C = 12.0 O = 16.0 Na = 23.0]

☒ **A** 106
☒ **B** 142
☒ **C** 263
☒ **D** 286

(Total for question = 1 mark)

Q17.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which aqueous solution contains the greatest number of **ions**?

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☒ **A** 200 cm^3 of 1.5 mol dm^{-3} $MgCl_2$
☒ **B** 400 cm^3 of 0.8 mol dm^{-3} $MgSO_4$
☒ **C** 500 cm^3 of 1.0 mol dm^{-3} NaCl
☒ **D** 1000 cm^3 of 0.25 mol dm^{-3} Na_2SO_4

(Total for question = 1 mark)

Q18.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A sample of seawater with a mass of 1 kg contains 6×10^{-9} g of gold.

How many atoms of gold, to one significant figure, are there in 1 g of this seawater?

[A_r value: Au = 197 Avogadro constant = 6×10^{23} mol^{-1}]

- ☐ **A** 2×10^{10}
- ☐ **B** 4×10^{12}
- ☐ **C** 2×10^{13}
- ☐ **D** 4×10^{15}

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(Total for question = 1 mark)

Q19.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The maximum permitted concentration of sulfur in diesel fuel is 10 mg of sulfur in 1 kg of diesel fuel.

(a) What is this concentration of sulfur in ppm?

(1)

- ☐ **A** 0.00001
- ☐ **B** 0.01
- ☐ **C** 10
- ☐ **D** 10 000

(b) 3.2 kg of this diesel fuel is burned in air.

What is the maximum volume, in dm^3, of sulfur dioxide which can be produced, measured at room temperature and pressure (r.t.p.)?
[Molar volume of a gas at r.t.p. = 24 dm^3 mol^{-1}]

(1)

- ☐ **A** 0.024
- ☐ **B** 0.77
- ☐ **C** 2.4
- ☐ **D** 24

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(Total for question = 2 marks)

Q20.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind

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about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A mass of 0.23 g of sodium was added to 350 cm^3 water to form hydrogen and a solution of sodium hydroxide.

$$Na(s) + H_2O(l) \rightarrow NaOH(aq) + \tfrac{1}{2}H_2(g)$$

(a) What is the concentration, in mol dm^{-3}, of sodium hydroxide in the solution formed?

(1)

- ☒ **A** 0.010
- ☒ **B** 0.029
- ☒ **C** 0.29
- ☒ **D** 0.66

(b) What is the maximum volume, in cm^3, of hydrogen which could be formed, measured at r.t.p.?

[Molar volume of a gas at r.t.p. = 24 dm^3 mol^{-1}]

(1)

- ☒ **A** 120
- ☒ **B** 240
- ☒ **C** 480
- ☒ **D** 2800

(c) The sodium hydroxide solution was neutralised with sulfuric acid.

Which is the ionic equation for this reaction?

(1)

- ☒ **A** $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$
- ☒ **B** $SO_4^{2-}(aq) + 2Na^+(aq) \rightarrow Na_2SO_4(aq)$
- ☒ **C** $H_2SO_4(aq) + 2Na^+(aq) + 2OH^-(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$
- ☒ **D** $2H^+(aq) + SO_4^{2-}(aq) + 2Na^+(aq) + 2OH^-(aq) \rightarrow 2Na^+(aq) + SO_4^{2-}(aq) + 2H_2O(l)$

(d) Sodium hydroxide solution was added to magnesium sulfate solution.

The equation for the reaction is shown.

$$2NaOH(aq) + MgSO_4(aq) \rightarrow Mg(OH)_2(s) + Na_2SO_4(aq)$$

What is the atom economy (by mass) for the production of magnesium hydroxide?

[A_r values: H = 1.0 O = 16.0 Na = 23.0 Mg = 24.3 S = 32.1]

(1)

- ☒ **A** 29.1 %
- ☒ **B** 41.0 %
- ☒ **C** 48.4 %
- ☒ **D** 50.0 %

(Total for question = 4 marks)

Q21.

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Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which aqueous solution has the **highest** concentration, in mol dm^{-3}, of chloride ions?

☐ **A** 0.1 g dm^{-3} HCl

☐ **B** 0.1 g dm^{-3} NaCl

☐ **C** 0.1 g dm^{-3} KCl

☐ **D** 0.1 g dm^{-3} $BaCl_2$

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(Total for question = 1 mark)

Q22.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An oxide of lead contains 90.7 % by mass of lead.

What is the formula of this oxide?

[A_r values: O = 16.0 Pb = 207.2]

☐ **A** PbO
☐ **B** PbO_2
☐ **C** Pb_2O_3
☐ **D** Pb_3O_4

(Total for question = 1 mark)

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Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these gases occupies 6.0 dm^3 at room temperature and pressure (r.t.p.)?

[molar volume of gas at r.t.p. = 24.0 dm^3 mol^{-1}

A_r values: He = 4.0 C = 12.0 N = 14.0 O = 16.0]

☐ **A** 2.0 g of helium
☐ **B** 4.0 g of oxygen
☐ **C** 11.0 g of carbon dioxide

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☒ **D** 14.0 g of nitrogen

(Total for question = 1 mark)

Q24.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

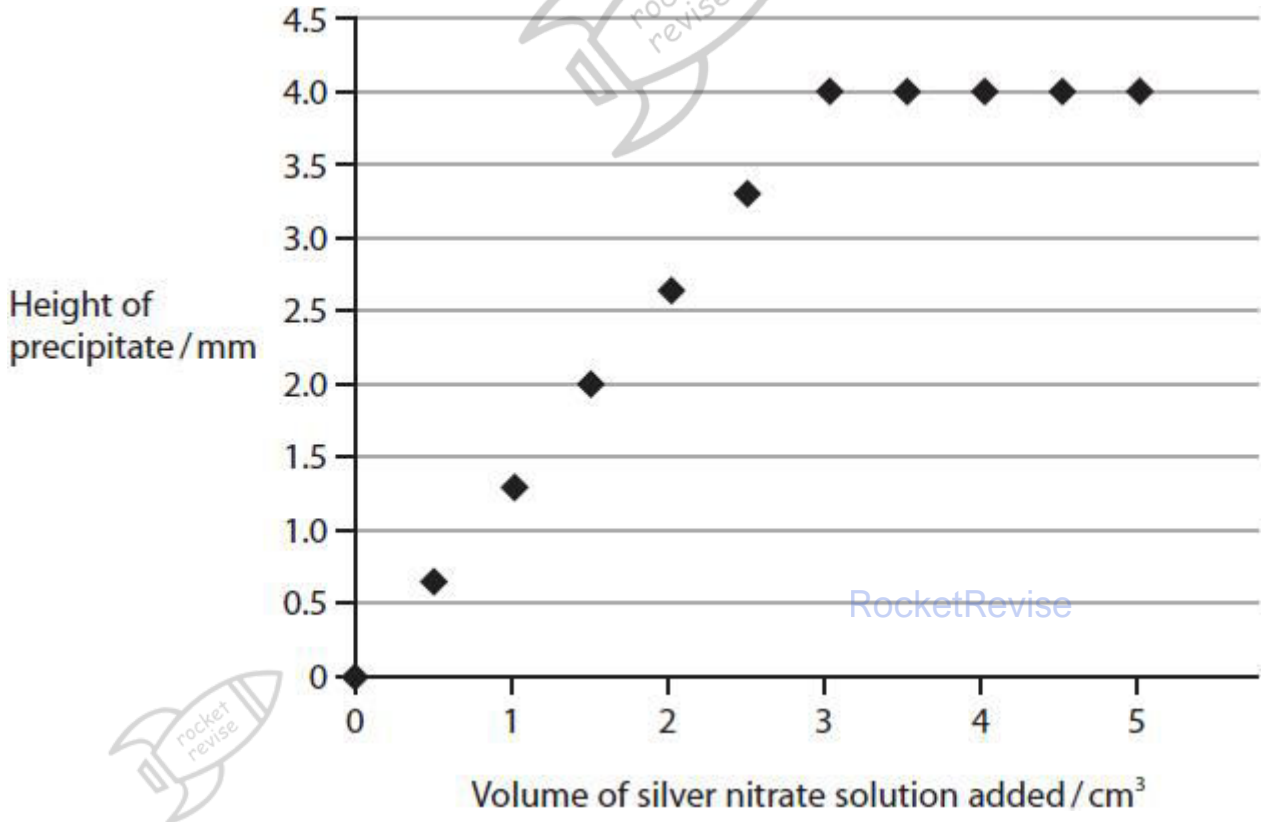
Ten test tubes, each containing 1.0 cm^3 of a chromium chloride solution of concentration 0.1 mol dm^{-3}, were placed in a test tube rack.

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Different volumes of silver nitrate solution of concentration 0.1 mol dm^{-3} were added to each test tube, giving a precipitate of silver chloride.

$$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$$

The precipitates formed were allowed to settle and their heights measured.
The results were plotted on a graph.



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What is the formula of the chromium chloride?

☒ **A** $CrCl$
☒ **B** Cr_3Cl_4
☒ **C** Cr_4Cl_3
☒ **D** $CrCl_3$

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(Total for question = 1 mark)

Q25.

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For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these solutions contains the greatest number of ions?

- ☒ **A** 20.0 cm^3 of 0.5 mol dm^{-3} KCl
- ☒ **B** 0.40 dm^3 of 0.03 mol dm^{-3} KCl
- ☒ **C** 10.0 cm^3 of 0.6 mol dm^{-3} $CaCl_2$
- ☒ **D** 0.15 dm^3 of 0.04 mol dm^{-3} $CaCl_2$

(Total for question = 1 mark)

Q26.

(a) The element iron forms two chlorides: iron(II) chloride, $FeCl_2$, and iron(III) chloride, $FeCl_3$.

A known mass of iron powder is added to 200 cm^3 of a hot solution of iron(III) chloride with a concentration of 0.500 mol dm^{-3}. When the reaction is complete, the solution only contains iron(II) chloride.

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The unreacted iron is filtered, dried and weighed.

Initial mass of iron powder = 6.17 g

Final mass of iron powder = 3.38 g

(i) Calculate the number of moles of iron that react.

(2)

(ii) Calculate the number of moles of iron(III) chloride that react.

(2)

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(iii) Use your answers to (a)(i) and (a)(ii) to write the ionic equation for the reaction of iron with iron(III) chloride. Include state symbols.
You **must** show your working.

(3)

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(b) The concentration of the solution obtained in (a) is increased by heating it gently to remove some of the water. The solution is allowed to cool and pale green crystals of a hydrated iron(II) chloride, FeCl2Â·xH2O, form.

Analysis shows that these crystals contain 28.1% by mass of iron.
Calculate the number of moles of water of crystallisation, x, per mole of hydrated iron(II) chloride.

(4)

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(Total for question = 11 marks)

Q27.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A sample of nitrogen gas contains 1.204×10^{22} molecules.

What is the mass of this sample?

[A_r N = 14.0 Avogadro constant (L) = 6.02×10^{23} mol^{-1}]

- ☒ **A** 0.14 g
- ☒ **B** 0.28 g
- ☒ **C** 0.56 g
- ☒ **D** 1.12 g

Use this space for any rough working. Anything you write in this space will gain no credit.

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(Total for question = 1 mark)

Q28.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The concentration of nitrogen dioxide in a sample of air is 0.5 ppm.

What is the percentage of nitrogen dioxide molecules in this sample of air?

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- ☒ **A** 0.5 %
- ☒ **B** 0.005 %
- ☒ **C** 0.00005 %
- ☒ **D** 0.0000005 %

(Total for question = 1 mark)

Q29.

This question is about the amount of water of crystallisation in hydrated magnesium sulfate, $MgSO_4 \cdot x$

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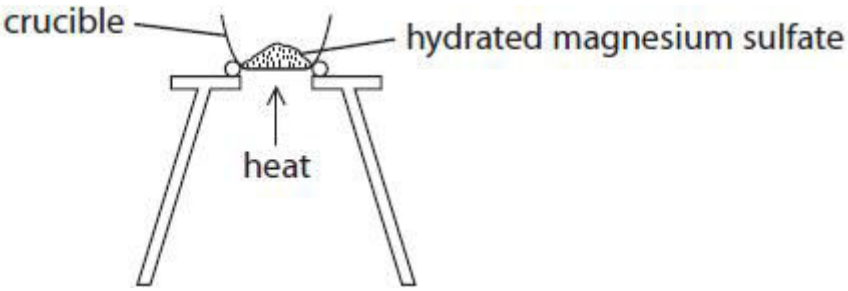
H_2O.

(a) The value of **x** in the formula was determined in an experiment.

Procedure

Step 1 A crucible was weighed, a spatula measure of hydrated magnesium sulfate was added and the crucible was reweighed.

Step 2 The crucible containing the hydrated magnesium sulfate was heated using the apparatus shown.



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Step 3 After heating for two minutes, the crucible containing the magnesium sulfate was allowed to cool and was reweighed.

(i) Complete the table of results.

(1)

Measurement	Mass / g
Mass of empty crucible	21.21
Mass of crucible and hydrated magnesium sulfate before heating	26.71
Mass of crucible and magnesium sulfate after heating for two minutes	24.12
Mass of magnesium sulfate after heating for two minutes	
Mass of water lost	

(ii) Use these results to calculate the value of **x** in $MgSO_4 \cdot \mathbf{x}\ H_2O$.

Give your answer to the nearest whole number.

[A_r values: H = 1.0 O = 16.0 Mg = 24.3 S = 32.1]

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(4)

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(b) The correct value of **x** is greater than the value calculated in (a)(ii).

Suggest a way of improving the method to obtain a more accurate result, using the same apparatus. Justify your answer.

(2)

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(Total for question = 7 marks)

Q30.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Potassium chlorate(V) decomposes on heating to form oxygen.

$$2KClO_3 \rightarrow 2KCl + 3O_2$$

What is the atom economy (by mass) for the formation of oxygen?

[A_r values: O = 16.0 Cl = 35.5 K = 39.1]

- ☐ **A** 13.1 %
- ☐ **B** 26.1 %
- ☐ **C** 39.2 %
- ☐ **D** 64.3 %

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(Total for question = 1 mark)

Q31.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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The equation for the complete combustion of hexane is shown.

$$C_6H_{14} + 9½O_2 \rightarrow 6CO_2 + 7H_2O$$

How many molecules of carbon dioxide are formed when 2×10^{-3} mol of hexane undergoes complete combustion?

[Avogadro constant $L = 6.02 \times 10^{23}$ mol^{-1}]

- ☒ **A** 1.20×10^{21}
- ☒ **B** 7.22×10^{21}
- ☒ **C** 8.43×10^{21}
- ☒ **D** 3.18×10^{23}

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(Total for question = 1 mark)

Q32.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Iron(III) oxide is reduced by hydrogen in a two-step process.

Step 1: $3Fe_2O_3 + H_2 \rightarrow 2Fe_3O_4 + H_2O$

Step 2: $Fe_3O_4 + 4H_2 \rightarrow 3Fe + 4H_2O$

What is the maximum mass of iron that could be produced from 39.9 tonnes of Fe_2O_3 ?

[A_r values: H = 1.0 O = 16.0 Fe = 55.8]

- ☒ **A** 6.98 tonnes
- ☒ **B** 13.95 tonnes
- ☒ **C** 27.90 tonnes
- ☒ **D** 41.85 tonnes

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(Total for question = 1 mark)

Q33.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A mass of 4.08 g of hydrated copper(II) sulfate, $CuSO_4 \cdot 5H_2O$, was dissolved in deionised water to form 250 cm^3 of solution.

What is the concentration of the copper(II) sulfate solution formed, in mol dm^{-3}?

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(1)

- ☒ **A** 0.0163
- ☒ **B** 0.0256
- ☒ **C** 0.0654
- ☒ **D** 0.102

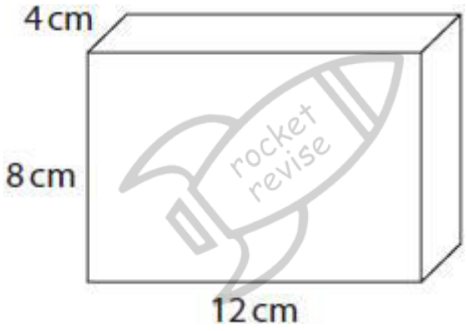
(Total for question = 1 mark)

Q34.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A piece of copper is shown.



How many moles of copper atoms are there in this piece of copper?

[Density of copper = 8.96 g cm^{-3}]

- ☒ **A** 0.675
- ☒ **B** 1.48
- ☒ **C** 6.05
- ☒ **D** 54.2

(Total for question = 1 mark)

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Q35.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A mass of 2.50 g of sodium chloride reacts with excess lead(II) nitrate solution forming lead(II) chloride with a
yield of 95 %.

What is the mass of lead(II) chloride, $PbCl_2$, formed?

$$2NaCl + Pb(NO_3)_2 \rightarrow PbCl_2 + 2NaNO_3$$

[M_r values: NaCl = 58.5 $PbCl_2$ = 278.2]

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- ☐ **A** 5.65 g
- ☐ **B** 5.94 g
- ☐ **C** 11.3 g
- ☐ **D** 11.9 g

(Total for question = 1 mark)

Q36.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A solution of sodium chloride, NaCl, is prepared by dissolving 10.0 g of the solid in distilled water and making the solution up to 250.0 cm^3.

What is the concentration of the solution, in mol dm^{-3} ?

[M_r NaCl = 58.5]

- ☐ **A** 0.171
- ☐ **B** 0.684
- ☐ **C** 10.0
- ☐ **D** 40.0

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(Total for question = 1 mark)

Q37.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

0.15 mol of aluminium is added to 120 cm^3 of 1.50 mol dm^{-3} hydrochloric acid.

$$2Al(s) + 6H^+(aq) \rightarrow 2Al^{3+}(aq) + 3H^2(g)$$

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The amount of **unused** reactant is

- ☐ **A** 0.06 mol aluminium
- ☐ **B** 0.09 mol aluminium
- ☐ **C** 0.03 mol hydrochloric acid
- ☐ **D** 0.13 mol hydrochloric acid

(Total for question = 1 mark)

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Q38.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which solution contains chloride ions with a concentration of 0.0500 mol dm^{-3}?

[A_r values: Ca = 40.1 Cl = 35.5 Na = 23.0]

	Solute	Mass of solute / g	Volume of solution / cm^3
☐ **A**	calcium chloride	1.39	250
☐ **B**	calcium chloride	1.39	500
☐ **C**	sodium chloride	1.46	250
☐ **D**	sodium chloride	1.46	1000

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(Total for question = 1 mark)

Q39.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 kg of seawater contains 64 mg of bromide ions.

[A_r of Br = 80 Avogadro constant, $L = 6.0 \times 10^{23}$ mol^{-1}]

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(a) What is the concentration of bromide ions in parts per million (ppm) by mass?

(1)

- ☐ **A** 0.80
- ☐ **B** 64
- ☐ **C** 800
- ☐ **D** 64 000

(b) How many bromide ions are in 500 g of the sample?

(1)

- ☐ **A** 2.4×10^{20}
- ☐ **B** 4.8×10^{20}
- ☐ **C** 1.9×10^{22}
- ☐ **D** 3.8×10^{22}

(Total for question = 2 marks)

Q40.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

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Concentrations in mixtures can be given in units of parts per million (ppm).

(a) A sample of a gaseous mixture contains 2.19 % carbon dioxide by mass.

What is the concentration of carbon dioxide in parts per million (ppm) by mass in this sample?

(1)

- ☐ **A** 2.19×10^{6}
- ☐ **B** 2.19×10^{4}
- ☐ **C** 2.19×10^{-4}
- ☐ **D** 2.19×10^{-6}

(b) A solution of lead nitrate has a concentration of 15 ppm by mass.

What is the mass of lead nitrate in 400 g of this solution?

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(1)

- ☒ **A** 6.00×10^{-3} g
- ☒ **B** 6.00×10^{-6} g
- ☒ **C** 6.00×10^{-3} kg
- ☒ **D** 6.00×10^{0} kg

(Total for question = 2 marks)

Q41.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the concentration of hydroxide ions in a solution containing 5.00 g of $Ba(OH)_2$ in 250 cm^3?

[M_r $Ba(OH)_2$ = 171.3]

- ☒ **A** 0.0292 mol dm^{-3}
- ☒ **B** 0.0584 mol dm^{-3}
- ☒ **C** 0.1168 mol dm^{-3}
- ☒ **D** 0.2335 mol dm^{-3}

(Total for question = 1 mark)

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Q42.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An organic compound contains 24.2 % carbon, 4.00 % hydrogen and 71.8 % chlorine by mass.

[A_r values: H = 1.0 C = 12.0 Cl = 35.5]

What is the empirical formula of the compound?

- ☒ **A** CHCl

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☒ **B** CH_2Cl
☒ **C** $C_2H_4Cl_2$
☒ **D** $C_4H_4Cl_4$

(Total for question = 1 mark)

Q43.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The formula of calcium carbide is CaC_2.

(a) What is the formula of the carbide ion?

(1)

☒ **A** C_2^-
☒ **B** C_2^+
☒ **C** C_2^{2-}
☒ **D** C_2^{2+}

(b) Excess calcium carbide and 10 g of water react to form the hydrocarbon ethyne, C_2H_2.

$$CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$$

What is the mass of ethyne that forms, assuming a yield of 100 %?
[M_r values: H_2O = 18.0 C_2H_2 = 26.0]

(1)

☒ **A** 7.22 g
☒ **B** 14.4 g
☒ **C** 23.4 g
☒ **D** 28.9 g

(Total for question = 2 marks)

Q44.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which compound contains 2.35×10^{22} molecules in 2.50 g?

[Avogadro constant $L = 6.02 \times 10^{23}$ mol^{-1}

M_r values: CO = 28.0 CO_2 = 44.0 SO_2 = 64.1 SO_3 = 80.1]

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- ☒ **A** CO
- ☒ **B** CO_2
- ☒ **C** SO_2
- ☒ **D** SO_3

(Total for question = 1 mark)

Q45.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A sample of 15.0 g of vanadium reacted completely with 7.07 g of oxygen.

What is the formula of the oxide?

[A_r values: V = 50.9 O = 16.0]

- ☒ **A** VO
- ☒ **B** V_2O_3
- ☒ **C** VO_2
- ☒ **D** V_2O_5

(Total for question = 1 mark)

Q46.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the pressure when 5.00 mol of nitrogen in a container with a volume of 1.50 m^3 is heated to 500 K?

[$pV = nRT$ $R = 8.31$ J K^{-1} mol^{-1}]

- ☒ **A** 13.9 Pa
- ☒ **B** 6 290 Pa
- ☒ **C** 13 900 Pa
- ☒ **D** 31 200 Pa

(Total for question = 1 mark)

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Q47.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The formula of phosgene is $COCl_2$.
What is the total number of **atoms** in 9.9 g of phosgene?

[Avogadro constant (L) = 6.02×10^{23} mol^{-1} M_r value: $COCl_2$ = 99.0]

- ☒ **A** 1.51×10^{22}
- ☒ **B** 6.02×10^{22}
- ☒ **C** 1.81×10^{23}
- ☒ **D** 2.41×10^{23}

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(Total for question = 1 mark)

Q48.

Which pair of substances contains the same number of moles at room temperature and pressure (r.t.p.)?

[A_r values Ca = 40, Li = 7, Al = 27, Mg = 24. Molar volume of gas at r.t.p. = 24 dm^3 mol^{-1}]

(1)

- ☒ **A** 24 dm^3 of chlorine, Cl_2, and 20 g of calcium, Ca
- ☒ **B** 24 dm^3 of oxygen, O_2, and 14 g of lithium, Li
- ☒ **C** 1.2 dm^3 of hydrogen, H_2, and 2.7 g of aluminium, Al
- ☒ **D** 1.2 dm^3 of nitrogen, N_2, and 1.2 g of magnesium, Mg

(Total for question = 1 mark)

Q49.

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A sample of a hydrocarbon with mass 7.2 g contained 6.0 g of carbon.

What is the empirical formula of the hydrocarbon?

(1)

- ☒ **A** CH_2
- ☒ **B** C_5H_{12}
- ☒ **C** C_6H_6
- ☒ **D** C_7H_6

(Total for question = 1 mark)

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Q50.

There is 0.045 g of solute in 1500 g of a solution.

What is the concentration of the solution in parts per million (ppm)?

(1)

- ☒ **A** 3.00
- ☒ **B** 6.75
- ☒ **C** 30.0
- ☒ **D** 67.5

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(Total for question = 1 mark)

Q51.

What is the concentration, in mol dm^{-3}, of a solution containing 7.84 g of phosphoric(V) acid, H_3PO_4, in 400 cm^3 of solution?

(1)

- ☒ **A** 0.02
- ☒ **B** 0.08
- ☒ **C** 0.20
- ☒ **D** 19.6

(Total for question = 1 mark)

Q52.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

This question is about the thermal decomposition of calcium nitrate.

$$2Ca(NO_3)_2 \rightarrow 2CaO + 4NO_2 + O_2$$

What volume of gas is produced by the complete decomposition of 0.050 mol of calcium nitrate at room temperature and pressure (r.t.p.)?

[Molar volume of a gas at r.t.p. = 24 dm^3 mol^{-1}]

- ☒ **A** 600 cm^3
- ☒ **B** 1.20 dm^3
- ☒ **C** 3.00 dm^3
- ☒ **D** 6.00 dm^3

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(Total for question = 1 mark)

Q53.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is the strongest **oxidising** agent in these displacement reactions?

$$Cl_2 + 2Br^- \rightarrow Br_2 + 2Cl^-$$

$$Br_2 + 2I^- \rightarrow 2Br^- + I_2$$

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- ☐ **A** chlorine
- ☐ **B** bromide ions
- ☐ **C** bromine
- ☐ **D** iodide ions

(Total for question = 1 mark)

Q54.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is the simplest ionic equation for an alkali reacting with an acid?

- ☐ **A** $2H^+(aq) + 2OH^-(aq) \rightarrow 2H_2O(l)$
- ☐ **B** $H^+(aq) + OH^-(aq) \rightarrow H_2O(aq)$
- ☐ **C** $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$
- ☐ **D** $2H^+(aq) + O^{2-}(aq) \rightarrow H_2O(l)$

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(Total for question = 1 mark)

Q55.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the maximum volume of butane that can be completely burned in 39 cm^3 of oxygen at room temperature and pressure?

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$$C_4H_{10}(g) + 6.5O_2(g) \rightarrow 4CO_2(g) + 5H_2O(l)$$

- ☒ **A** 6.0 cm^3
- ☒ **B** 12 cm^3
- ☒ **C** 24 cm^3
- ☒ **D** 45 cm^3

(Total for question = 1 mark)

Q56.

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Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Copper metal can displace silver from silver nitrate solution according to the equation shown.

$$Cu + 2AgNO_3 \rightarrow 2Ag + Cu(NO_3)_2$$

10 g of copper metal was added to an excess of silver nitrate solution.

The silver metal was collected, washed with deionised water and left to dry.

What is the mass of silver metal collected, assuming a 100% yield?

- ☒ **A** between 10 g and 20 g
- ☒ **B** 20 g
- ☒ **C** between 20 g and 40 g
- ☒ **D** more than 40 g

(Total for question = 1 mark)

Q57.

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Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Ethanol is formed from glucose during fermentation.

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$$

What is the atom economy by mass for the formation of ethanol in this reaction?

- ☒ **A** 26%
- ☒ **B** 49%
- ☒ **C** 51%
- ☒ **D** 96%

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(Total for question = 1 mark)

Q58.

Sodium hydroxide can be obtained as a hydrate, $NaOH \cdot xH_2O$.

When heated, the water of crystallisation is lost, leaving anhydrous sodium hydroxide, NaOH, as shown in the equation.

$$NaOH \cdot xH_2O \rightarrow NaOH + xH_2O$$

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An experiment was carried out to determine the value of x in $NaOH \cdot xH_2O$.

Procedure

Step **1** Weigh and record the mass of a clean, dry crucible.

Step **2** Add approximately 1.0 g of $NaOH \cdot xH_2O$ to the crucible and record the mass.

Step **3** Heat the crucible and its contents until a constant mass has been reached.

Step **4** After allowing to cool, reweigh the crucible and the anhydrous solid.

Step **5** Calculate and record the mass of the anhydrous solid.

Repeat Steps **1** to **5** using a different mass of the hydrated sodium hydroxide.

Results

Mass of $NaOH \cdot xH_2O$ / g	Mass of NaOH / g
1.00	0.69
2.10	1.45
3.50	2.41
4.90	3.38
6.60	4.55
8.00	5.52

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(a) (i) Complete a graph of the results by

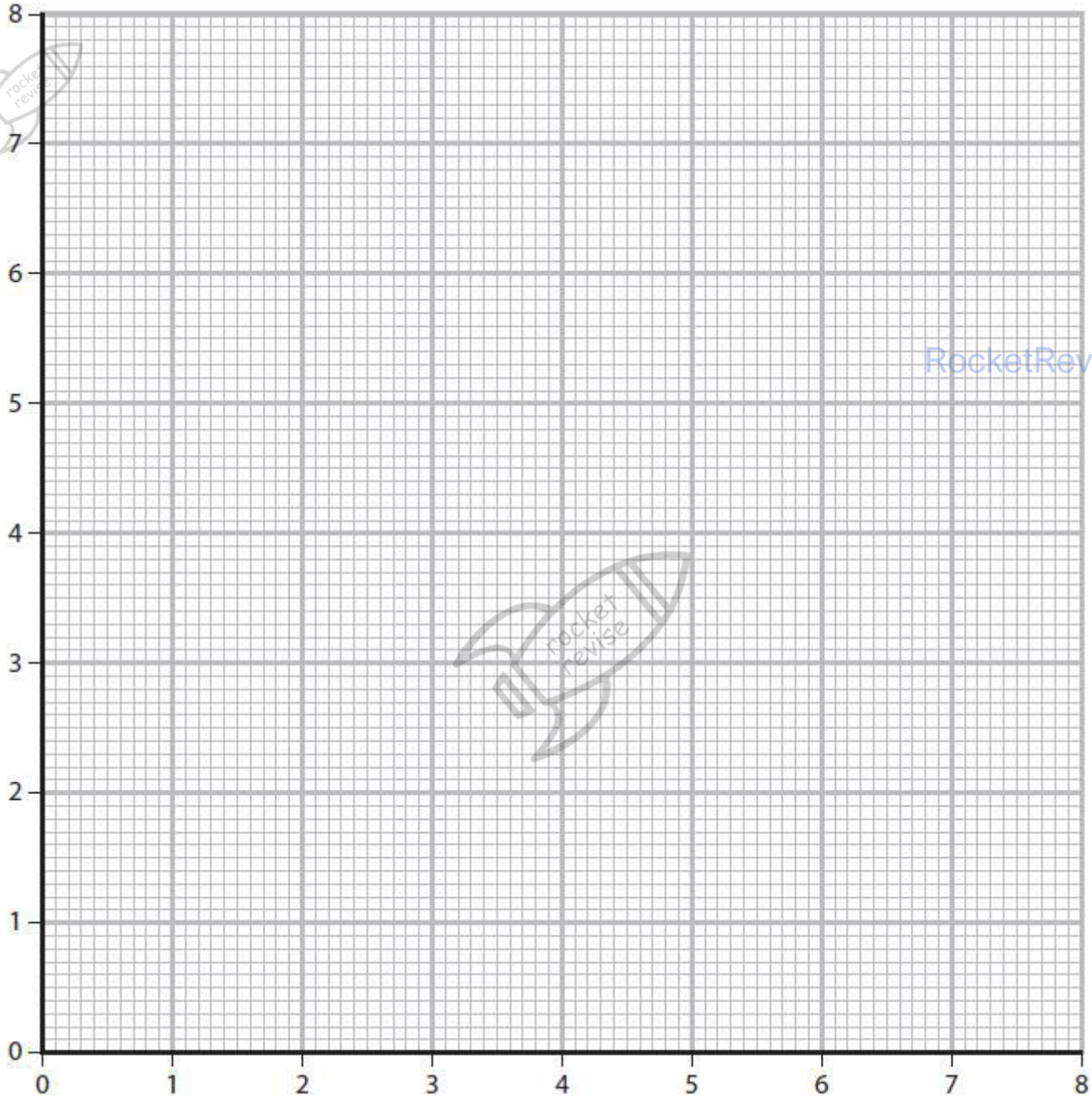
- plotting the points
- labelling the axes

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- including a straight line of best fit.

(3)



(ii) Use your graph to determine the mass of $NaOH \cdot xH_2O$ needed to form 4.0 g of NaOH. You must show your working on the graph.

(1)

.....

(iii) Calculate the value of x in $NaOH \cdot xH_2O$ using your answer to (a)(ii) and the equation for the reaction.

$$NaOH \cdot xH_2O \rightarrow NaOH + xH_2O$$

(3)

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(b) Sodium hydroxide also forms a heptahydrate, $NaOH \cdot 7H_2O$.

Calculate the mass of this heptahydrate needed to make 250 cm^3 of a solution of sodium hydroxide of concentration 0.150 mol dm^{-3}.

(2)

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(Total for question = 9 marks)

Q59.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which pair of compounds will **not** react to form a precipitate?

- ☐ **A** $AgNO_3(aq)$ and $CaCl_2(aq)$
- ☐ **B** $Ca(OH)_2(aq)$ and $CO_2(g)$
- ☐ **C** $Ba(OH)_2(aq)$ and $HCl(aq)$
- ☐ **D** $KI(aq)$ and $Pb(NO_3)_2(aq)$

(Total for question = 1 mark)

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Q60.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is observed when iron(II) carbonate reacts with ethanoic acid?

- ☐ **A** colourless solution
- ☐ **B** colourless solution and effervescence
- ☐ **C** green solution and effervescence
- ☐ **D** green solution

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(Total for question = 1 mark)

Q61.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Diphosphane, P_2H_4, reacts spontaneously with oxygen.

$$xP_2H_4 + yO_2 \rightarrow P_4O_{10} + zH_2O$$

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The equation for this reaction is balanced when

- ☐ **A** x = 1 y = 6 z = 2
- ☐ **B** x = 2 y = 6 z = 2
- ☐ **C** x = 2 y = 7 z = 4
- ☐ **D** x = 4 y = 9 z = 8

(Total for question = 1 mark)

Q62.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

How many oxygen **atoms** are there in 0.0100 mol of H_2SO_4?

[Avogadro constant, $L = 6.020 \times 10^{23}$ mol^{-1}]

- ☐ **A** 6.020×10^{21}
- ☐ **B** 1.204×10^{22}
- ☐ **C** 2.408×10^{22}
- ☐ **D** 4.214×10^{22}

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(Total for question = 1 mark)

Q63.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which oxide of nitrogen contains 30% nitrogen by mass?

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[A_r values: N = 14.0 O = 16.0]

- ☐ **A** NO
- ☐ **B** NO_2
- ☐ **C** N_2O
- ☐ **D** N_2O_3

(Total for question = 1 mark)

Q64.

RocketRevise

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The concentration of sulfur dioxide in a sample of polluted air is 0.4 ppm.

What is the percentage of sulfur dioxide molecules in this polluted air?

- ☐ **A** 0.4%
- ☐ **B** 0.004%
- ☐ **C** 0.00004%
- ☐ **D** 0.0000004%

(Total for question = 1 mark)

Q65.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A sample of seaweed contains 30.0 mg of iodine per kg.

What is the number of iodine **atoms** in 10 kg of this seaweed?

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[A_r value: I = 126.9 Avogadro constant $L = 6.02 \times 10^{23}$ mol^{-1}]

- ☐ **A** 7.12×10^{19}
- ☐ **B** 1.42×10^{20}
- ☐ **C** 7.12×10^{20}
- ☐ **D** 1.42×10^{21}

(Total for question = 1 mark)

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Q66.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An excess of sodium sulfate solution is added to 50 cm^3 of a 0.100 mol dm^{-3} solution of barium chloride.

What is the mass of barium sulfate formed?

[M_r value: $BaSO_4$ = 233.4]

$$Na_2SO_4(aq) + BaCl_2(aq) \rightarrow 2NaCl(aq) + BaSO_4(s)$$

- ☐ **A** 1.167 g
- ☐ **B** 2.334 g
- ☐ **C** 11.67 g
- ☐ **D** 23.34 g

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(Total for question = 1 mark)

Q67.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Two identical sealed flasks, containing different gases, are side by side. Each flask contains one gas, with the gases at the same temperature and pressure.

Flask **A** contains 4.0×10^{-3} mol of methane.

Flask **B** contains 160 mg of a different gas.

Which could be the gas in Flask **B**?

- ☐ **A** argon
- ☐ **B** carbon dioxide
- ☐ **C** helium
- ☐ **D** neon

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(Total for question = 1 mark)

Q68.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

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Magnesium azide is an ionic compound containing the azide ion, N_3^-.

The empirical formula of magnesium azide is

- ☒ **A** Mg_3N
- ☒ **B** MgN_2
- ☒ **C** MgN_3
- ☒ **D** MgN_6

(Total for question = 1 mark)

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Q69.

The density of an unknown gas is 0.656 g dm^{-3} at 20°C and 101 000 Pa.

[$pV = nRT$ $R = 8.31$ J mol^{-1} K^{-1}]

(a) Calculate the molar mass of the unknown gas.

(5)

(b) The unknown gas is a hydrocarbon.

Give the name or formula for the unknown gas using your answer to (a).

(1)

.....

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(Total for question = 6 marks)

Q70.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Calculate the mass of sodium carbonate (Na_2CO_3) required to make up 250 cm^3 of a 0.100 mol dm^{-3} solution.

[A_r values: C = 12.0 O = 16.0 Na = 23.0]

- ☒ **A** 1.30 g

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☒ **B** 2.65 g
☒ **C** 5.30 g
☒ **D** 10.6 g

(Total for question = 1 mark)

Q71.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A block of lead measuring 10 cm × 10 cm × 10 cm contains 3.295×10^{25} atoms.

Calculate the density of lead.

[A_r value: Pb = 207.2 Avogadro constant, $L = 6.02 \times 10^{23}$ mol^{-1}]

☒ **A** 3.79 g cm^{-3}
☒ **B** 4.49 g cm^{-3}
☒ **C** 11.34 g cm^{-3}
☒ **D** 54.73 g cm^{-3}

(Total for question = 1 mark)

Q72.

Baking powder is added to cake mixtures to make cakes 'rise' by releasing a gas during cooking.

Baking powder contains an acidic derivative of tartaric acid and about 30% by mass of sodium hydrogencarbonate.

Baking powder releases the gas in two types of reaction during cooking, one of which is neutralisation.

(a) (i) Give the name of the second type of reaction.

(1)

.....

(ii) Suggest why neutralisation does not occur in an unopened container of baking powder.

(1)

.....

.....

(b) (i) The acidic derivative of tartaric acid contains 20.8% potassium, 25.5% carbon, 2.66% hydrogen by mass and the rest is oxygen.

Calculate the empirical formula.

(2)

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(ii) The acidic derivative of tartaric acid can be represented by the formula $H^+K^+A^{2-}$.
Write the equation for the neutralisation of sodium hydrogencarbonate by $H^+K^+A^{2-}$.
State symbols are not required.

(2)

(iii) Calculate the maximum volume, in **cm^3**, of carbon dioxide gas released by the neutralisation reaction in (b)(ii) from 5.00 g of baking powder, in an oven at 190°C at 101 000 Pa.
Baking powder contains 30% sodium hydrogencarbonate by mass.
[pV = nRT]

(5)

(iv) Calculate the decrease in volume when the carbon dioxide gas is cooled to 20°C.

(2)

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(Total for question = 13 marks)

Q73.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

How many **atoms** are there in 10 g of methane?

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[A_r values: H = 1.0 C = 12.0 $L = 6.02 \times 10^{23}$ mol^{-1}]

- ☐ **A** 3.76×10^{23}
- ☐ **B** 9.63×10^{23}
- ☐ **C** 1.88×10^{24}
- ☐ **D** 4.82×10^{24}

(Total for question = 1 mark)

Q74.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A solution contains 95 ppm of a solute.

What mass of the solute is dissolved in 1 kg of this solution?

- ☐ **A** 95 g
- ☐ **B** 0.95 g
- ☐ **C** 0.095 g
- ☐ **D** 0.000095 g

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(Total for question = 1 mark)

Q75.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the total number of **atoms** in 8.8 g of carbon dioxide?

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[M_r value: CO_2 = 44 Avogadro constant, L = 6.02 × 10^{23} mol^{-1}]

- ☒ **A** 1.2 × 10^{23}
- ☒ **B** 3.6 × 10^{23}
- ☒ **C** 1.2 × 10^{24}
- ☒ **D** 3.6 × 10^{24}

(Total for question = 1 mark)

Q76.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒

How many moles of **ions** are there in 40 cm^3 of a solution of 0.050 mol dm^{-3} magnesium nitrate, $Mg(NO_3)_2$?

- ☒ **A** 0.0020 mol
- ☒ **B** 0.0040 mol
- ☒ **C** 0.0060 mol
- ☒ **D** 0.0080 mol

(Total for question = 1 mark)

Q77.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Four different solutions contain chloride ions.
Three of the solutions contain the same number of moles of chloride ions.

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Which solution contains a different number?

- ☒ **A** 15.0 cm^3 of 0.80 mol dm^{-3} hydrochloric acid
- ☒ **B** 10.0 cm^3 of 0.40 mol dm^{-3} iron(III) chloride solution
- ☒ **C** 10.0 cm^3 of 0.90 mol dm^{-3} magnesium chloride solution
- ☒ **D** 20.0 cm^3 of 0.60 mol dm^{-3} sodium chloride solution

(Total for question = 1 mark)

Q78.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

N_xO_4 is an oxide of nitrogen.
The percentage by mass of oxygen in this oxide is 69.57%.

What is the relative molecular mass of this oxide?

- ☒ **A** 78
- ☒ **B** 92
- ☒ **C** 106
- ☒ **D** 109

(Total for question = 1 mark)

Q79.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Ethanoic acid can be produced by the oxidation of butane.

$$2C_4H_{10} + 5O_2 \rightarrow 4CH_3COOH + 2H_2O$$

The atom economy, by mass, for the production of ethanoic acid is

- ☒ **A** 13%
- ☒ **B** 67%
- ☒ **C** 77%
- ☒ **D** 87%

(Total for question = 1 mark)

Q80.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A water molecule, containing the isotopes of hydrogen and oxygen, hydrogen-1 and oxygen-16, has a mass of 2.9908×10^{-23} g.

What is the mass of an atomic mass unit (amu) measured in grams?

[A_r values: oxygen-16 = 15.995 amu hydrogen-1 = 1.0080 amu]

- ☒ **A** 6.0185×10^{23}
- ☒ **B** 6.0221×10^{23}
- ☒ **C** 1.6605×10^{-24}

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☒ **D** 1.6616×10^{-24}

(Total for question = 1 mark)

Q81.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

How many molecules of H_2O are present in 1 drop (0.050 cm^3) of water?

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[$L = 6.02 \times 10^{23}$, density of water = 1 g cm^{-3}]

☒ **A** 2.17×10^{26}
☒ **B** 6.69×10^{23}
☒ **C** 1.67×10^{22}
☒ **D** 1.67×10^{21}

(Total for question = 1 mark)

Q82.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the maximum volume of butane which will completely burn in 130 cm^3 of oxygen?

$$C_4H_{10} + 6.5O_2 \rightarrow 4CO_2 + 5H_2O$$

☒ **A** 10 cm^3
☒ **B** 15 cm^3
☒ **C** 20 cm^3
☒ **D** 40 cm^3

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(Total for question = 1 mark)

Q83.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which mass of fertiliser contains 467 g of nitrogen?

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- ☐ **A** 2 kg sodium nitrate ($NaNO_3$; M_r = 85)
- ☐ **B** 1 kg urea (NH_2CONH_2; M_r = 60)
- ☐ **C** 1 kg ammonium nitrate (NH_4NO_3; M_r = 80)
- ☐ **D** 1 kg ammonium sulfate ($(NH_4)_2SO_4$; M_r = 132.1)

(Total for question = 1 mark)

Q84.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A sample of hydrated magnesium sulfate contains 43% by mass of water.

What is the formula of this magnesium sulfate?

- ☐ **A** $MgSO_4 \cdot 9H_2O$
- ☐ **B** $MgSO_4 \cdot 5H_2O$
- ☐ **C** $MgSO_4 \cdot 2.5H_2O$
- ☐ **D** $MgSO_4 \cdot 1.25H_2O$

(Total for question = 1 mark)

Q85.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What volume of sulfur dioxide gas reacts completely with 50 cm^3 of 0.12 mol dm^{-3} sodium hydroxide solution?

[molar volume of a gas = 24 dm^3 at room temperature and pressure]

$$SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$$

- ☐ **A** 0.072 dm^3
- ☐ **B** 0.144 dm^3
- ☐ **C** 0.288 dm^3
- ☐ **D** 72 dm^3

(Total for question = 1 mark)

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Q86.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A 3.50 g sample of an organic compound will completely burn forming 5.15 g of carbon dioxide and 2.11 g of water.

What is the empirical formula of the compound?

- ☐ **A** CH_2O
- ☐ **B** CH_3O
- ☐ **C** C_2HO
- ☐ **D** CH_2O_2

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(Total for question = 1 mark)

Q87.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which reaction to produce copper(II) nitrate (M_r = 187.5) has an atom economy of 46.5% by mass?

- ☐ **A** $Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$
- ☐ **B** $2Cu + Pt(NO_3)_4 \rightarrow 2Cu(NO_3)_2 + Pt$
- ☐ **C** $Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$
- ☐ **D** $CuO + 2HNO_3 \rightarrow Cu(NO_3)_2 + H_2O$

(Total for question = 1 mark)

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Q88.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which method for the preparation of hydrogen has the highest atom economy by mass?

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☐ **A** $2H_2O \rightarrow 2H_2 + O_2$

☐ **B** $CH_4 + H_2O \rightarrow 3H_2 + CO$

☐ **C** $CO + H_2O \rightarrow H_2 + CO_2$

☐ **D** $CH_3OH + H_2O \rightarrow 3H_2 + CO_2$

(Total for question = 1 mark)

Q89.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which amount does **not** have a mass of approximately 6 g?

☐ **A** 0.33 mol water (H_2O; M_r = 18)

☐ **B** 0.25 mol magnesium (Mg; A_r = 24.3)

☐ **C** 0.033 mol glucose ($C_6H_{12}O_6$; M_r = 180)

☐ **D** 0.025 mol copper(II) sulfate ($CuSO_4$; M_r = 159.6)

(Total for question = 1 mark)

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